

CIFAR-100 Speedrun



This challenge is reasonably hands off, so we advise you to tackle it in parallel with one of the other challenges at the hackathon!

Welcome! This track is organised by C3 and AIDDA. C3 provides GPU and CPU compute for research groups and the agents that work with them, while AIDDA coordinates research, knowledge sharing and networking around AI-driven algorithm discovery. We're excited to see what you build.

At a glance

- Experiment: Develop a faster way to train an image classifier for CIFAR-100. Autoresearch tools are encouraged, and you can choose any approach.
- Submit: Open a pull request with your submission in `submissions/my_team`.
- Deadline: 4 October, **2:45pm**.
- Judging: The lowest average training time wins, provided average test accuracy is at least 75% across 40 trials.
- Resources: Compute and LLM/API tokens are provided for the hackathon.
- Support: Join the event Discord <https://discord.gg/ZN36RsFPs>, find us at the event, or email admin@algorithmdiscovery.org.
- Prize: \$1000 of compute credits!

Repository: <https://github.com/AIDDA-Institute/CIFAR-100-speedrun>

The challenge

How quickly can you train an accurate image classifier?

CIFAR-100 contains 60,000 colour images, each 32 by 32 pixels: 50,000 training images and 10,000 test images across 100 categories. These include apples, bears, bicycles, dolphins, lions, rockets and tractors.

A training speedrun has a simple objective: reach a fixed level of accuracy in as little training time as possible. You can improve the model architecture, optimiser, training schedule, data augmentation or the code that runs it.

Inspired by training speedruns such as CIFAR-10 and NanoGPT, this track is a chance to help establish a new CIFAR-100 benchmark.

Getting started

1. Fork the repository and follow the README to set up the environment and run the starter example.
2. Copy `submission_template` into `submissions/my_team`, replacing `my_team` with your team name.
3. Develop your model and training recipe. You are encouraged to use autoresearch tools to suggest ideas, run experiments and improve code.
4. Test your changes with a small number of trials while developing, then check promising recipes across more seeds.
5. Submit your final recipe through a pull request before the deadline.

The README and submission guide explain the three functions you implement: `build` sets up reusable resources, `prepare` starts each trial fresh, and `train` teaches the model. The benchmark runner handles timing and accuracy checks.

You can use any development or autoresearch tools, but the submitted code must run in the repository's pinned PyTorch environment. Train

only on the CIFAR-100 training split; pretrained weights, external training datasets and learned state carried between trials are not allowed.

Read the full rules before you start:

<https://github.com/AIDDA-Institute/CIFAR-100-speedrun/blob/main/RULES.md>

What you submit

Submit the source code for your model and training algorithm, with any supporting source files, in submissions/my_team. Your pull request should add only your team's folder. Your submission should work with its default settings and train a fresh model when the judges run it. Do not submit trained weights or checkpoints.

You should not include the autoresearch system used to develop your submission.

Judging

Each submission is evaluated on one NVIDIA A100 GPU using the repository's fixed software environment. Every submission will use the same 40 random seeds. A seed controls random choices such as the initial model weights and the order of training examples.

Each trial starts training from scratch. After training, the model is evaluated on all 10,000 test images. For each image, its highest-scoring category is its prediction; accuracy is the percentage of predictions that are correct.

All 40 trials must complete successfully. Judges will calculate:

- Mean test accuracy across the 40 trials. This must be at least 75% to qualify; individual trials may fall below 75%.
- Mean preparation plus training time across the 40 trials. The qualifying submission with the lowest mean time wins.

Inference does not count towards the training-time score, but the complete test-set evaluation must finish within 5 seconds per trial.

The repository's RULES.md contains the full timing rules, resource limits and submission requirements.

Compute and LLM access

Each team will receive compute resources and LLM/API tokens for the duration of the hackathon iterate.inc/london-ai-science?tab=resources.

The instructions will explain which compute resources and models are available, the usage limits, and how to obtain your credentials.

Contact us

Please reach out if you need help during the event.

Discord: <https://discord.gg/ZN36RsFPs>

Email: admin@algorithmdiscovery.org